Ajara Oyinloye Lawal

Data Scientist

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Profile

Data scientist finishing a master's at Carnegie Mellon University Africa, with hands-on work across computer vision, LLM agents and time-series forecasting. I like the whole loop: cleaning and labelling data, building and debugging models, and designing the evaluation that shows whether they really work. I have a software engineering background in Python, FastAPI and PostgreSQL, and I care about building AI that holds up in African and low-resource settings.

Education

MS in Engineering Artificial Intelligence 2025 – Dec 2026
Carnegie Mellon University Africa, Kigali, Rwanda

- **Coursework:** Deep Learning, Applied Computer Vision, Machine Learning, Natural Language Processing, AI Systems Design, Statistics and Data Inference

BSc Computer Science 2019 – 2023
University for Development Studies, Ghana

Experience

Summer Intern, RISE Programme Summer 2026
Carnegie Mellon University Africa, Kigali

- Completed facilitation training as part of the programme

Student Affairs Support 2025 – Present
Carnegie Mellon University Africa, Kigali

- Manage student communications, calendars and administrative workflows using Google Workspace
- Coordinate campus event logistics, vendor procurement and budget records in Excel and Google Sheets

Junior Software Developer (Co-op) 2023 – 2024
Ghana-India Kofi Annan Centre of Excellence in ICT, Accra, Ghana

- Co-developed two enterprise systems, SmartConference and a Staff Appraisal ERP, with Python FastAPI and PostgreSQL, automating internal workflows
- Tested and validated backend APIs with Swagger to keep the systems reliable
- Taught Python, office productivity and data analytics to students and working professionals

Operations Manager (Intern) Nov 2021 – Mar 2025
Namibra, Ghana

Selected Projects

Lightweight Human Detection for Search-and-Rescue Drones 2026

- Fine-tuned YOLOv8n/s and RetinaNet on aerial SAR imagery with a two-layer augmentation pipeline, reaching 94.31% mAP@0.5 at 178.5 FPS, 7.5 points above a published baseline with a model 20x smaller
- Combined YOLOv8n with Depth Anything V2 and a custom polynomial calibration to estimate distance to each person, with 2.04 m mean absolute error over 1 to 24.5 m

Autonomous IT Support Agent 2025

- Built a three-agent system (triage, diagnosis, resolution) on Llama 3.3 70B with retrieval from Stack Overflow and GitHub Issues, completing 100% of tasks across 1,000+ university IT queries with 87.5% tool-selection accuracy
- Designed an evaluation framework with five automated metrics (groundedness, coherence, tool accuracy, completion, confidence) and used its error analysis to cut false resolutions from 30% to 0%
- Added persistent SQLite memory so the agent learns across sessions, raising confidence on recurring issues by 3.7%

Object Detection and Segmentation Benchmark 2026

- Compared YOLOv8, Faster R-CNN, EfficientDet and RetinaNet on a dataset our team collected and annotated in Roboflow; diagnosed and fixed a RetinaNet run returning zero predictions, then ran controlled experiments to reach 0.936 mAP@0.5:0.95
- Benchmarked SAM, CLIPSeg, U-Net, Mask R-CNN, DeepLabV3 and SegFormer for object counting; a U-Net trained from scratch beat the pretrained models, and classic Watershed beat them all

Explainable AI Early Warning for Floods and Droughts in Africa 2025

- Built preprocessing pipelines for large environmental datasets in Python, NumPy and Pandas, and contributed to an explainable climate-prediction model
- Co-authored a research paper on AI for disaster prevention in low-resource settings

Forecasting International Homicide Rates 2025

- Compared LSTM networks in PyTorch with statistical time-series methods, evaluating with RMSE and error analysis, and visualised results for policy audiences

Ask Your Data: Agentic BI Prototype 2026

- Designed and built an interactive prototype that lets non-technical business users ask questions about company finances in plain English and get source-linked answers, with generated SQL available on demand

Robustness of Medical AI Models 2026, ongoing

- Working in a team of four to test how medical models hold up under adversarial attacks, with careful handling of a small dataset

Skills

- **Programming:** Python, SQL, JavaScript, HTML, CSS, PHP
- **Machine learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, CNNs, LSTMs
- **Computer vision:** YOLOv8, RetinaNet, Faster R-CNN, U-Net, SAM, Depth Anything V2, Roboflow annotation, COCO evaluation
- **LLMs:** LLM agents, retrieval-augmented generation, multi-agent systems, evaluation design
- **Data and tools:** Pandas, NumPy, Matplotlib, FastAPI, PostgreSQL, SQLite, Git, Linux, Kaggle, Google Colab, Excel, Google Workspace
- **Ways of working:** Agile and Scrum, design thinking

Leadership and Community

Treasurer, CMU-Africa Data Science Club Dec 2025 – Dec 2026

- Plan event budgets, secure sponsorship and outside support, co-organise events and track club activities

Country Representative, Ghana, CMU Africa Dec 2025 – Dec 2026

- Represent Ghanaian students, welcome new arrivals, organise hangouts and support country, multicultural and diversity days

Community and Volunteering

- Organised community sessions teaching young learners coding, game development, animation and internet safety
- As a CAMFED alumna, volunteer to teach professional skills and mentor girls in basic education
- Earlier roles as Society Secretary, Audit Committee Member and Procurement Officer

Certificate

Virtual Assistant Programme, ALX Africa